

This assignment implements an image description and search system following the steps presented in **Lecture 3**. The system was developed and tested on **CSC Puhti** using a GPU interactive session.

System overview

The pipeline processes a collection of images and builds a searchable “model” consisting of:

1. The image filename
2. A textual description generated by a vision–language model
3. A numerical embedding vector computed from the description

For image understanding, the **ministral-3:3b** model was used via the **Ollama** framework. For semantic representation, text embeddings were computed using the **embeddinggemma** model. All embeddings are stored as NumPy arrays and organized in a Pandas DataFrame.

Model construction

All images in the dataset (including bear.jpg and multiple images from the frames/ folder) were processed one by one. For each image, a short English description was generated and then converted into an embedding vector. These results were combined into a DataFrame with columns filename, description, and embedding. The model was saved to disk as a CSV file and can be reloaded for later use.

Image search

To perform image search, a user query (for example, “*Show me a bear*”) is first embedded using the same embedding model. Cosine similarity is then computed between the query embedding and all stored image embeddings. The image with the highest similarity score is returned as the best match.

Demonstration

The final step (“Serving frontend”) was implemented directly in a Jupyter notebook, as recommended in the assignment instructions. A text query was submitted, the system returned the most relevant image filename and description, and the corresponding image was displayed in the notebook. A screenshot of this result is provided as part of the submission.

Summary

The assignment demonstrates an end-to-end image description and retrieval system, including batch processing, model storage, similarity search, and interactive querying. All required tasks from the lecture slides were completed, and the system behaves as expected on the test queries.

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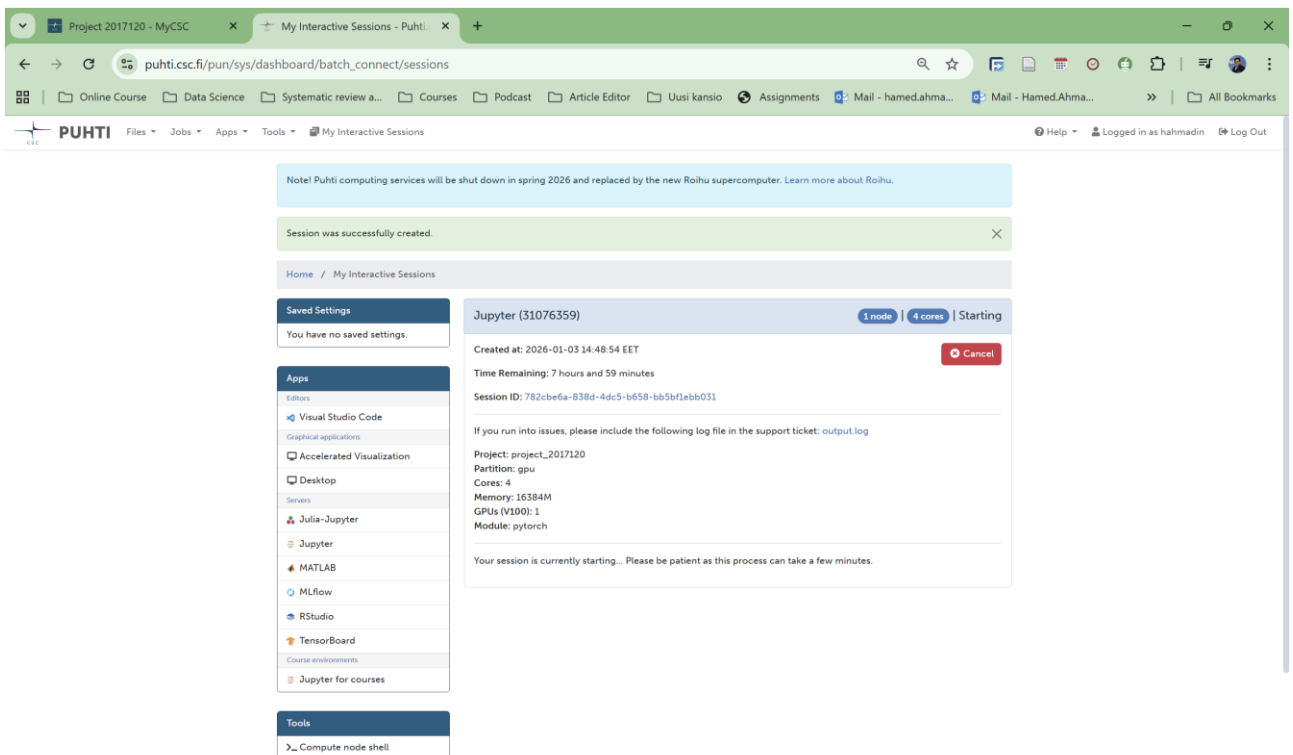
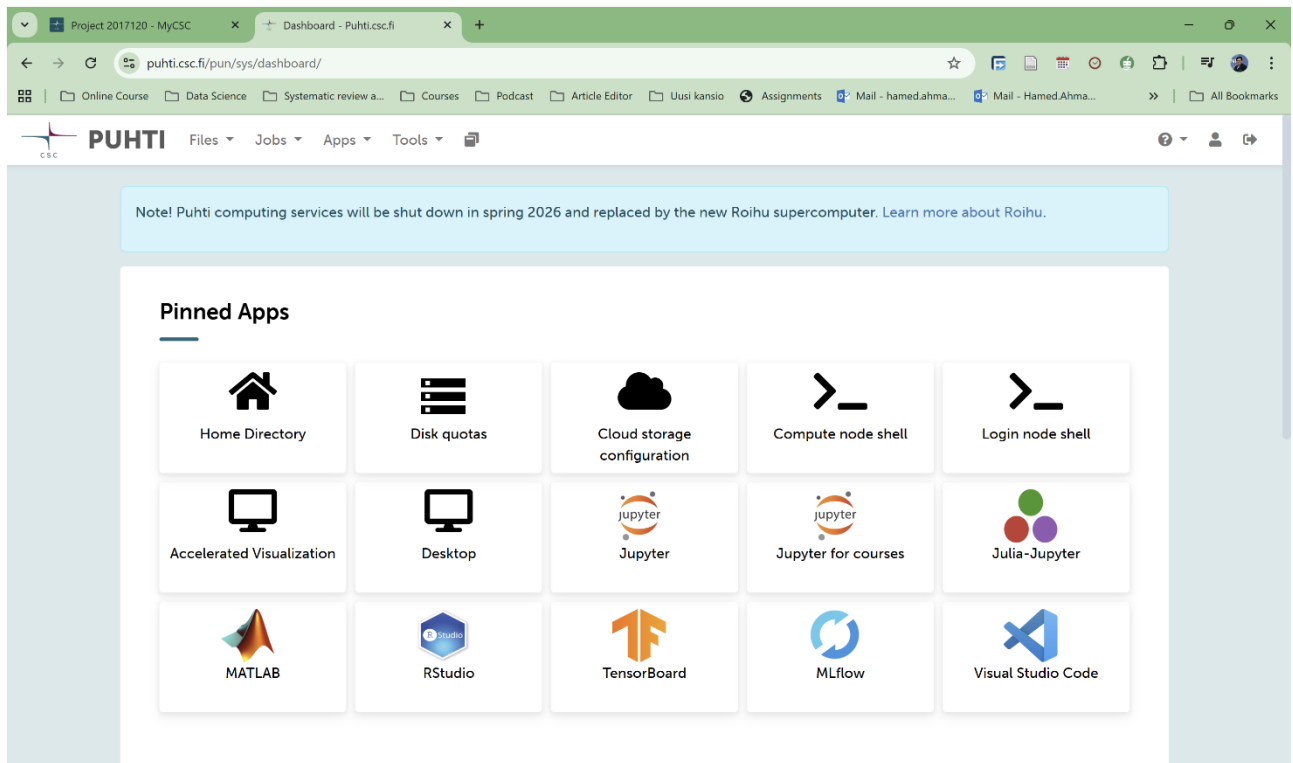
FilterGenerate invitation linkAdd members

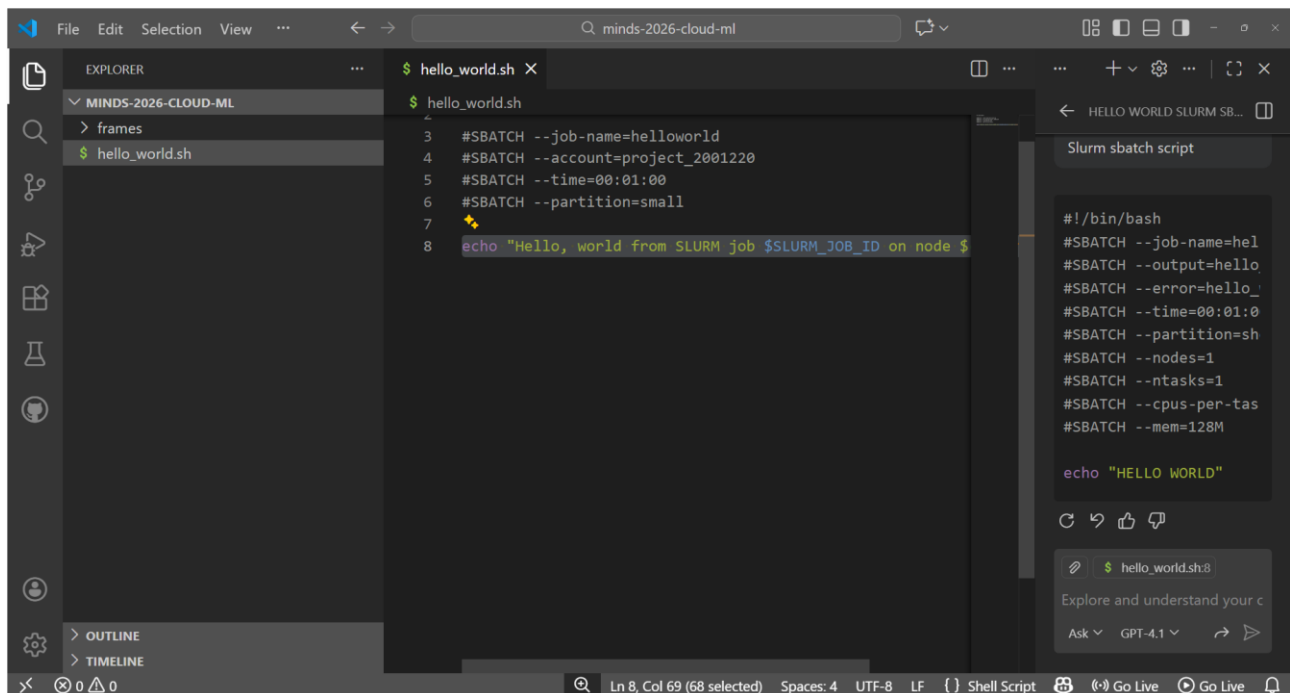
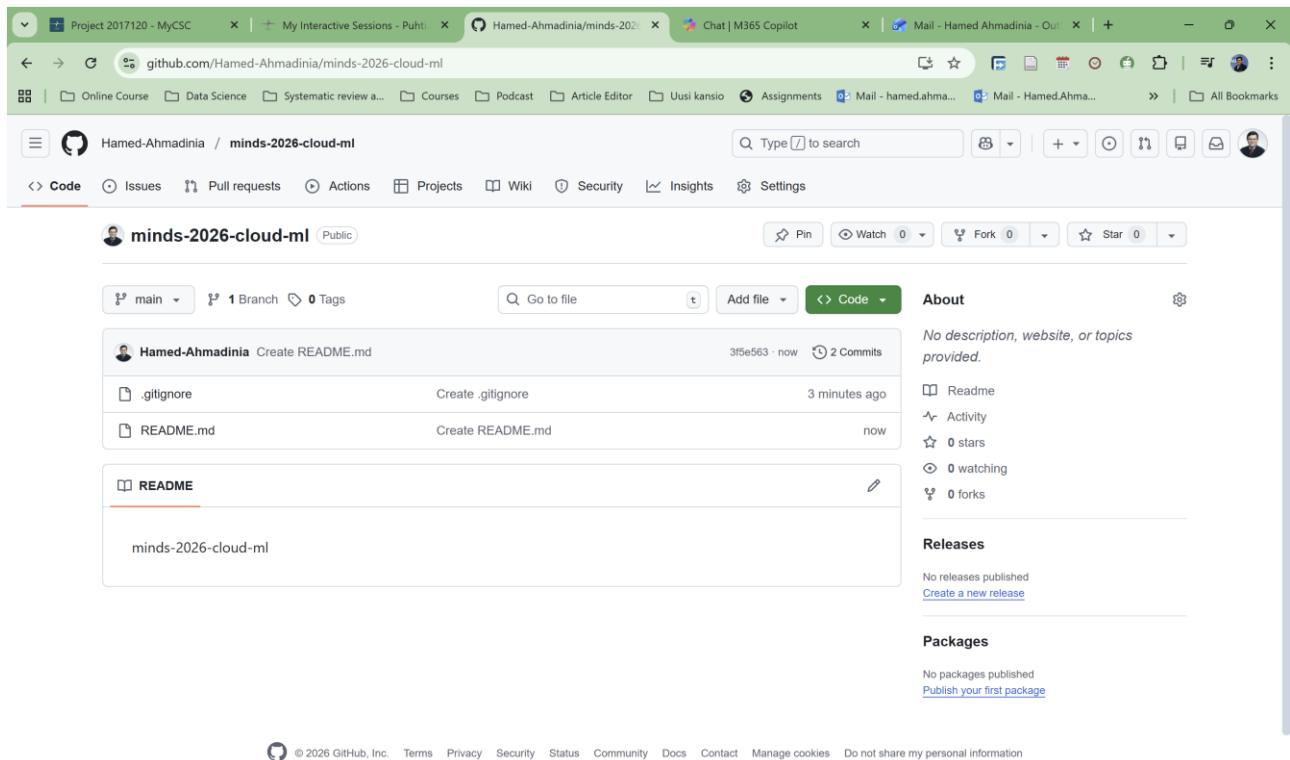
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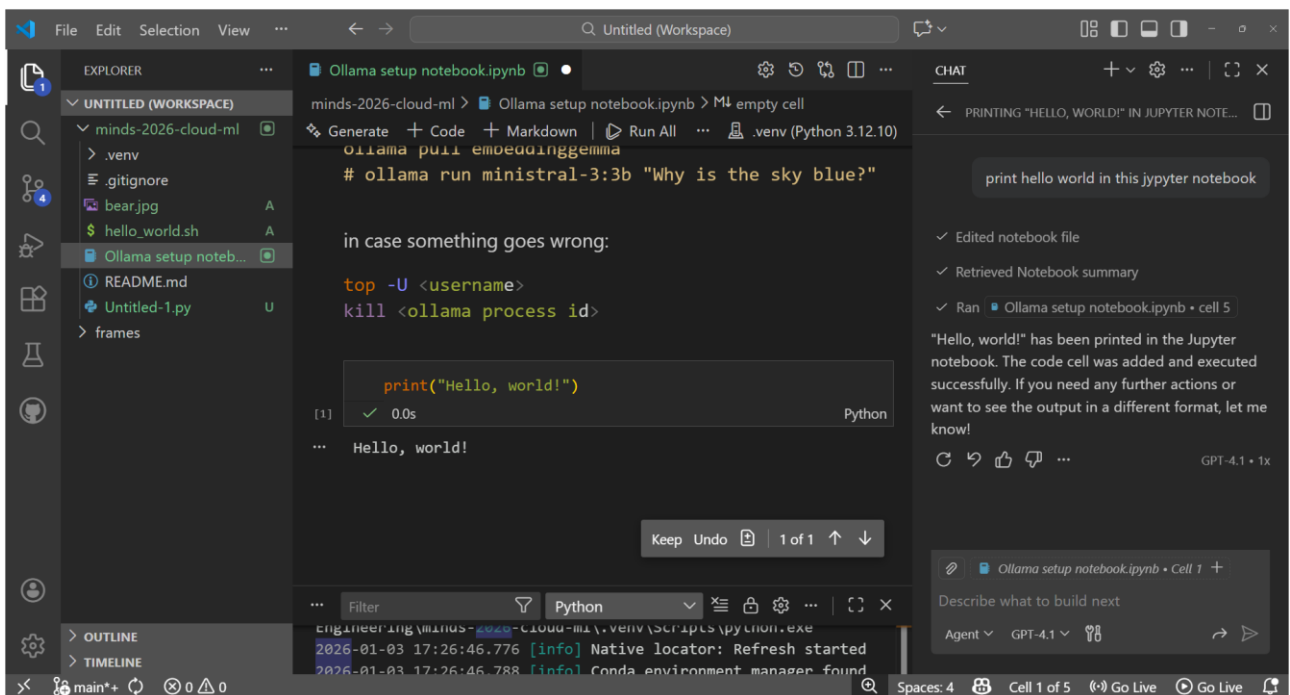
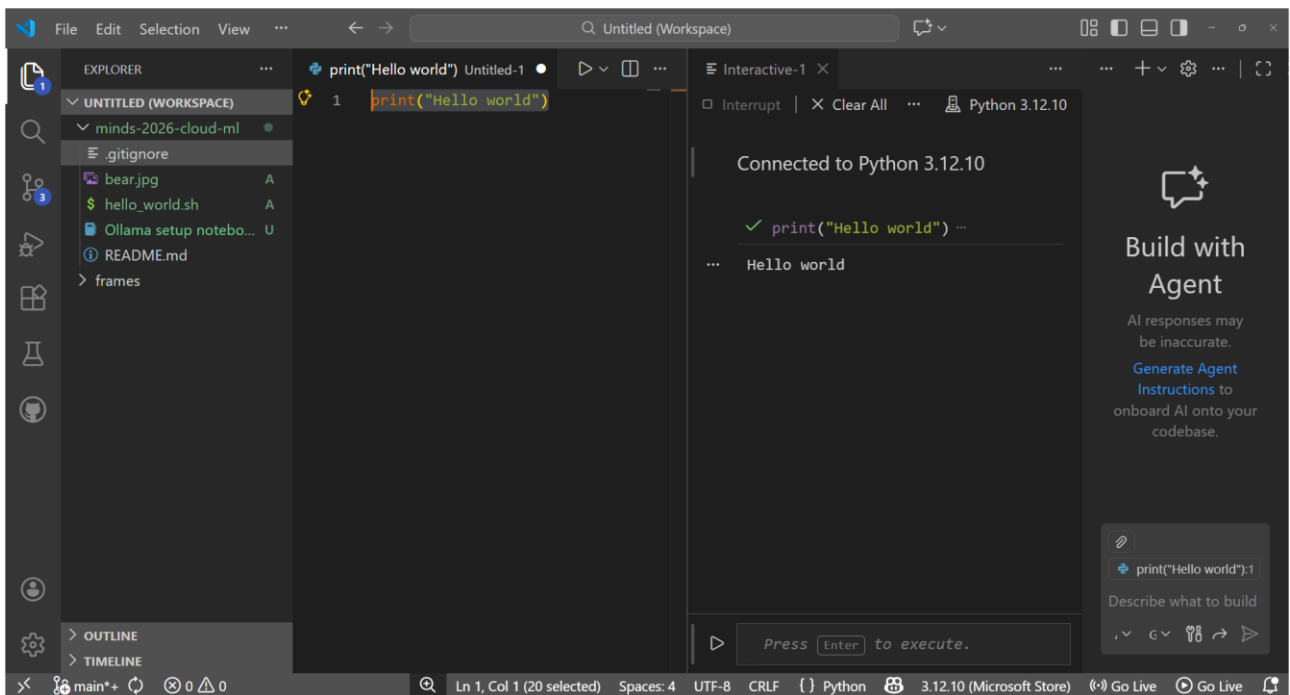
Username ↑

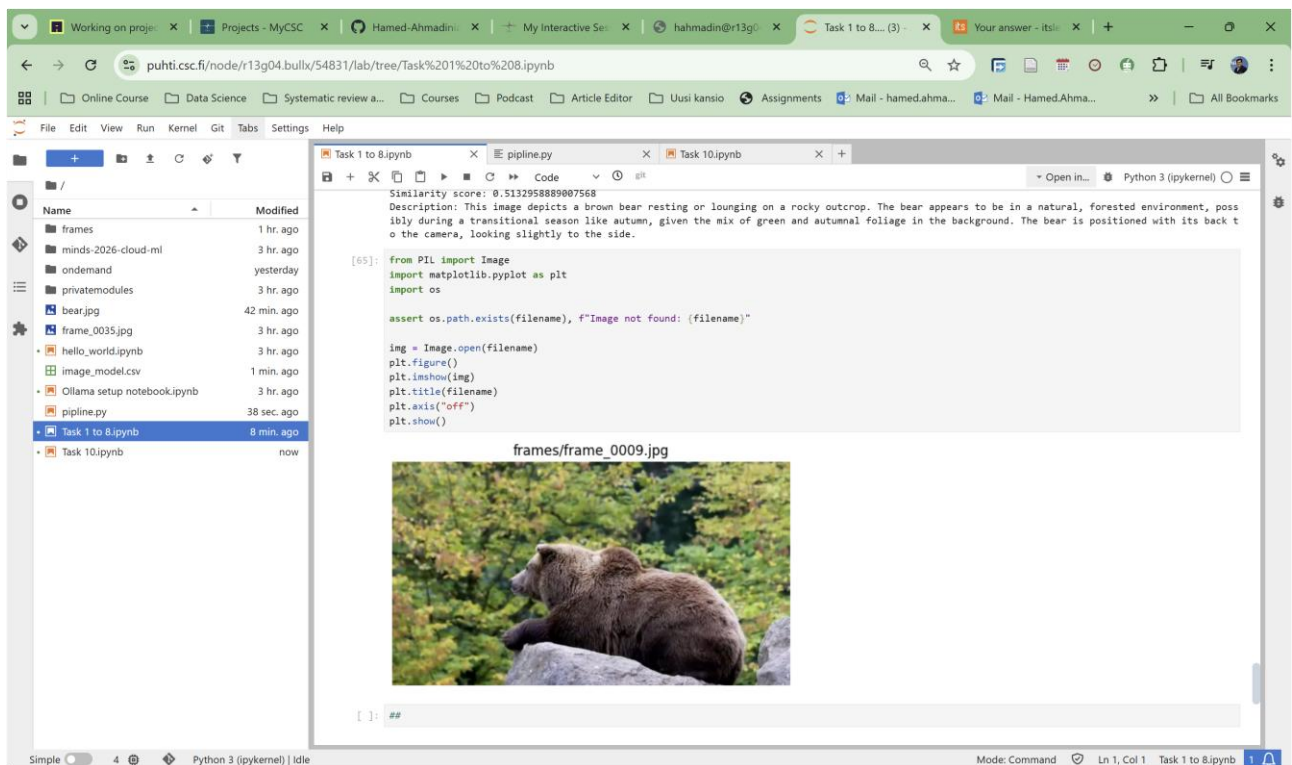
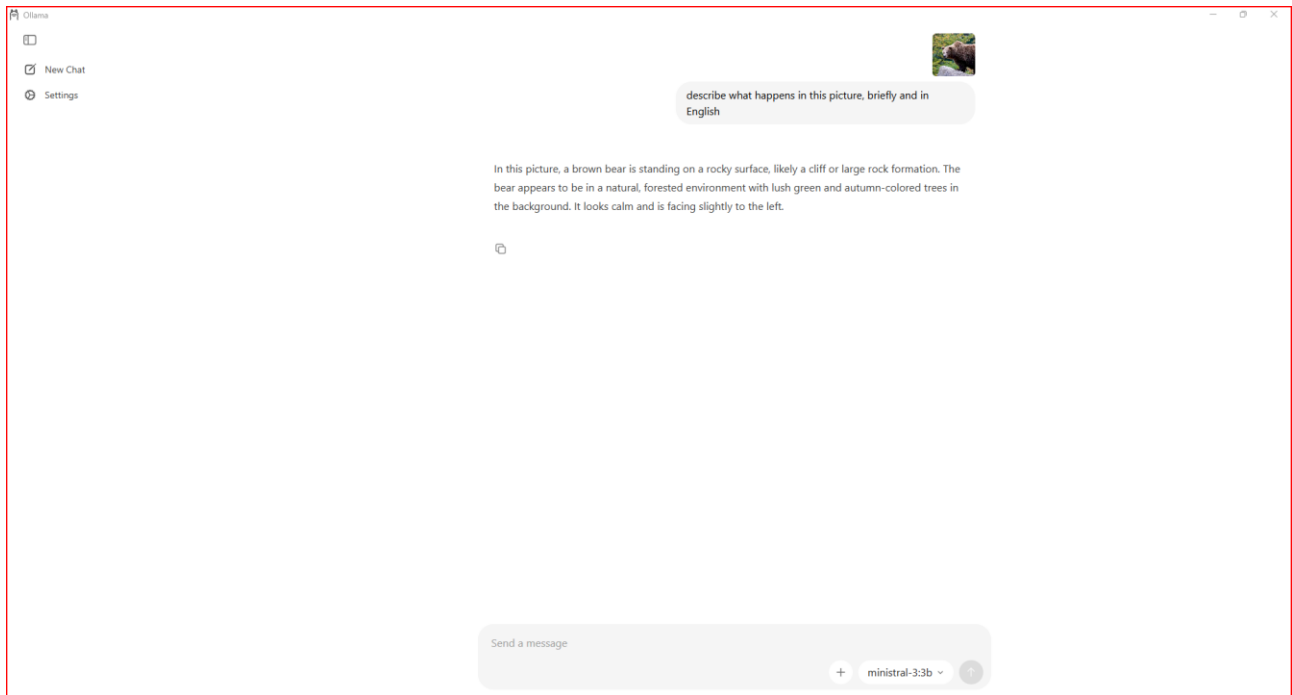
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Python 3 (ipykernel)

```
# retrieve best match
filename, description, score = find_most_similar(df_model, q_emb)

# show results
print("Query:", query)
print("Best match:", filename, "score:", score)
print("Best description:", description)

# show image
img_path = os.path.join(image_dir, filename)
if os.path.exists(img_path):
    display(Image.open(img_path))
else:
    print("WARNING: Image file not found at:", img_path)

return filename, description, score
```

```
[6]: serve_query("Show me a bear", model_csv="image_model.csv", image_dir="frames")

Query: Show me a bear
Best match: frames/frame_0009.jpg score: 0.5132958292961121
Best description: This image depicts a brown bear resting or lounging on a rocky outcrop. The bear appears to be in a natural, forested environment, possibly during a transitional season like autumn, given the mix of green and autumnal foliage in the background. The bear is positioned with its back to the camera, looking slightly to the side.
WARNING: Image file not found at: frames/frames/frame_0009.jpg

[6]: ('frames/frame_0009.jpg',
      'This image depicts a brown bear resting or lounging on a rocky outcrop. The bear appears to be in a natural, forested environment, possibly during a transitional season like autumn, given the mix of green and autumnal foliage in the background. The bear is positioned with its back to the camera, looking slightly to the side.',
      0.5132958292961121)

[ ]:
```

Simple 4 Python 3 (ipykernel) | Idle Mode: Edit Ln 13, Col 52 Task 10.ipynb